Analytic parameter sensitivities in ManifoldsX: validation and a robot case study

manifoldsx.github.io/differentiable-dynamics

ManifoldsX computes how motion and joint loads change with spring and bushing stiffness and damping, analytically, at every step of a simulated trajectory. The derivatives cover consistent initialization, the implicit index-3 HHT step and the rotation update. They were compared with independent perturbed simulations over 30-second runs. A case study then uses them to identify the seven joint stiffnesses of a robot arm from one 8-second tool recording, on simulated data with tracker noise. The checks are numerical; they say nothing about hardware accuracy.

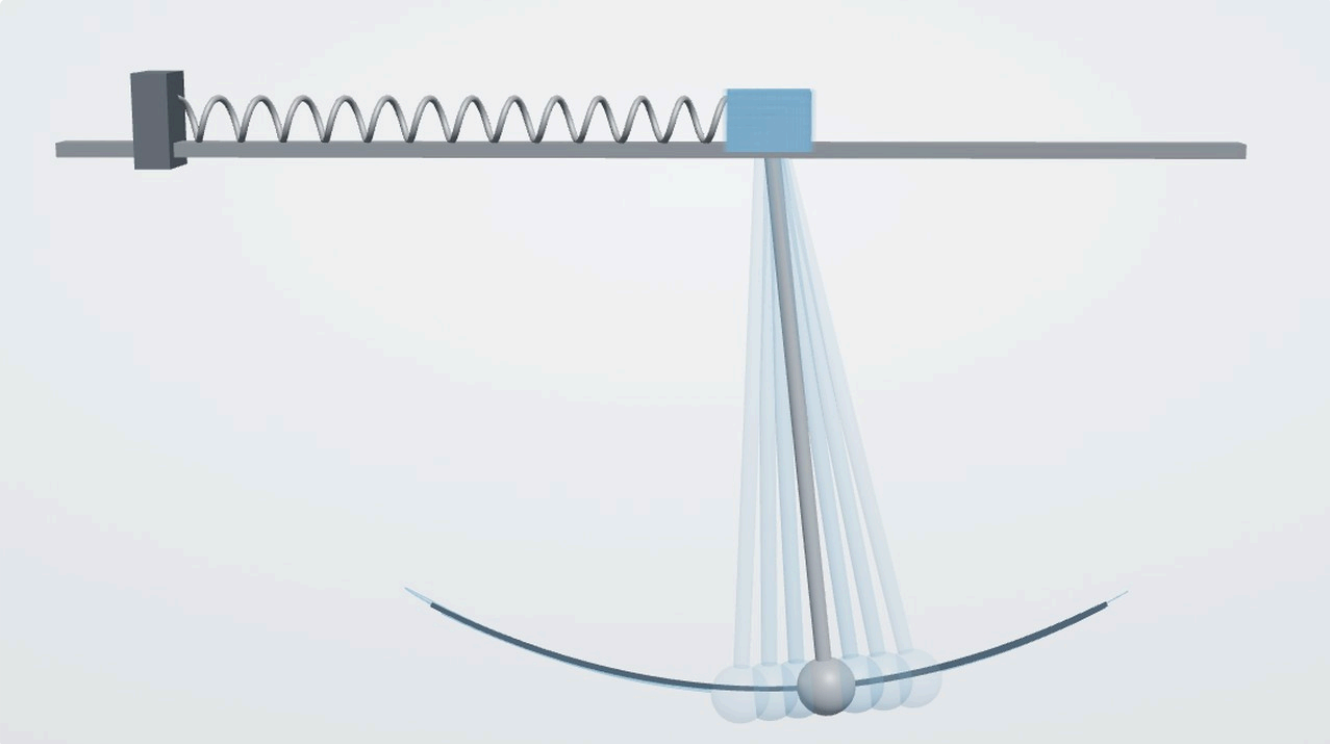


Figure 1. The slider-pendulum-spring test model. The solid pendulum is the simulated motion; the faint copies show how it shifts when the spring stiffness changes, predicted from the analytic gradients of a single run.

Background

“What if this part were stiffer?” is one of the oldest questions in machine design, and the usual way to answer it is to change the number and run the simulation again. Computing the derivative directly is the more efficient route. Forward sensitivity equations follow the derivative of the state alongside the state itself, while adjoint methods run backwards from a quantity of interest. Both are well established for differential-algebraic equations in numerical analysis [1]. They are also what

makes a simulator “differentiable”: calibration, design optimization and learning methods can follow a gradient instead of guessing.

For a model of a mechanism the derivative has to pass through everything the solver does in a step. That includes the consistent initial state, the implicit index-3 HHT solve with its Newton iterations [2], [3], and the update of rotations, which do not add like ordinary vectors. Finite differences avoid that work, but they need an extra simulation for every parameter, and their accuracy depends on a perturbation size that has to be guessed. Here they serve only as the reference, extrapolated to remove most of their own error.

1. What they are for

Understanding compliance

How a stiffness or damping coefficient changes displacement, velocity, acceleration and reaction forces at every instant.

Calibration and design

Gradients are the building blocks for parameter fitting and design studies. Section 3 shows a complete fit on simulated data; a fit to a real machine still has to be checked against its measurements.

2. Validation

Table 1. Validation results

Study	Evidence
Spring stiffness and damping	Analytic trajectory derivatives compared with Richardson-extrapolated finite differences over 30-second runs; every closed-form term also checked against finite differences.
6D bushings	All 12 diagonal coefficients; 24 trajectories of 30 seconds on grounded and moving-parent connections.
Bushing motion derivatives	Worst sampled error relative to the peak reference: 1.11×10^{-7} for position and rotation, 2.37×10^{-7} for velocity.
Bushing accelerations and loads	Same metric: 3.80×10^{-5} for acceleration, 5.47×10^{-5} for multipliers and 5.46×10^{-5} for generalized reactions. These are limited by the finite-difference reference, not by the derivatives.
Rotary motion drivers	Constraint-reaction derivative compared with Richardson-extrapolated finite differences at random states, with fixed and moving parents: worst relative error 6×10^{-11}. Robot-arm trajectory derivatives (Section 3) agree with central differences to 5×10^{-6}, the truncation level of the differences.
Sparse sensitivity solve	Damped chains from 10 to 500 links, with one symbolic pattern analysis per run.

Errors are normalized by the peak reference value over the sampled times, not per component at every instant. The analytic runs used no finite-difference evaluations; independent perturbed simulations supplied the reference. In the bushing tests, computing sensitivities left the nominal trajectory bit-for-bit unchanged.

3. Case study: joint stiffness of a robot arm

Serial robots repeat a motion well but place a tool less accurately than a machine tool, largely because their joints twist under load. Gears, bearings and torque sensors all add compliance, and the result is a tool that sits millimetres from where the controller puts it. For machining and drilling this is a known obstacle [4], [7]. Robot makers rarely publish joint stiffness, so it has to be identified, usually from static tests that load the arm in several poses and measure the tool [5]. A static test has a blind spot: the base joint turns about a vertical axis, so gravity puts no torque on it. Only inertial loads, from a moving arm, reveal its stiffness. Dynamic identification with periodic excitation is well established for inertial parameters [6]; it needs a model whose response to every parameter is known along the whole motion, which is what the sensitivities supply.

Model and test

The arm has seven revolute joints with the modified Denavit–Hartenberg layout of a common collaborative arm, 15.5 kg of links and a 3 kg tool whose centre of mass sits 0.1 m off the flange axis. Each joint is a motor side driven by a prescribed angle, an output link on a revolute joint, and a torsional bushing between them with stiffness k_i and damping c_i about the joint axis. Damping is taken as known. The true stiffnesses run from 24 to 3.6 kN m/rad, and the static tool sag is about 3 mm.

The arm first settles under gravity for 1.5 s with twenty times its joint damping, a known schedule that leaves the derivatives exact, and then holds still. From 2 to 10 s every joint moves at once: a sine of 0.4 to 0.8 rad at 0.21 to 0.47 Hz, faded in over 0.8 s. The “measurement” is the tool point every 10 ms from a run at a 0.25 ms step, with independent 20 µm noise on each axis: 801 samples. The fitted model uses a 1 ms step; the two steps differ by about 10^{-7} m.

Fit

Gauss–Newton works on joint compliance $1/k$, because tool deflection is nearly linear in compliance: a linear prediction from the true point misses a 553 µm change by 5 µm. Each iteration is one simulation carrying seven sensitivities, one per joint. From catalogue values 25 to 178% too stiff, the misfit fell from 554 µm to 19.8 µm after one iteration and 19.7 µm after the second, which is the noise level. The whole fit took three gradient runs and one plain run.

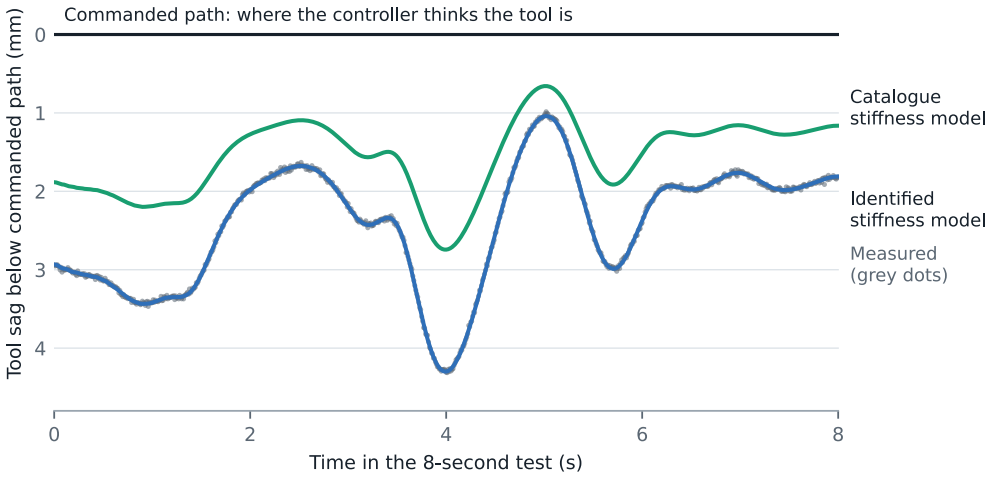


Figure 2. Tool sag below the commanded path during the test: tracker data (grey), the model with catalogue stiffness (green) and with identified stiffness (blue).

Table 2. Identified joint stiffness. Error is identified minus true, relative to true; 1σ is from the Gauss–Newton covariance with the known noise.

Joint	True (kN m/rad)	Catalogue start	Identified error	1σ
1 (base)	24	+25%	+0.2%	1.1%
2	19	+58%	+0.1%	0.1%
3	14	+43%	+0.1%	0.2%
4	13	+54%	0.0%	0.1%
5	7	+43%	−0.7%	1.0%
6	6	+67%	−0.3%	0.5%
7	3.6	+178%	−2.8%	1.3%

All seven errors are within the scatter expected from the noise; with noise-free data the same fit returns every stiffness within 5×10^{-3}%. The largest correlation between two estimates is 0.71. The base joint, invisible to a static test, comes out within 0.2%. Joint 7 is the least certain because the tool puts the smallest torque on it; an earlier tool with its centre of mass 0.04 m off axis gave it a 6% uncertainty, which is why the tool here is mounted to one side.

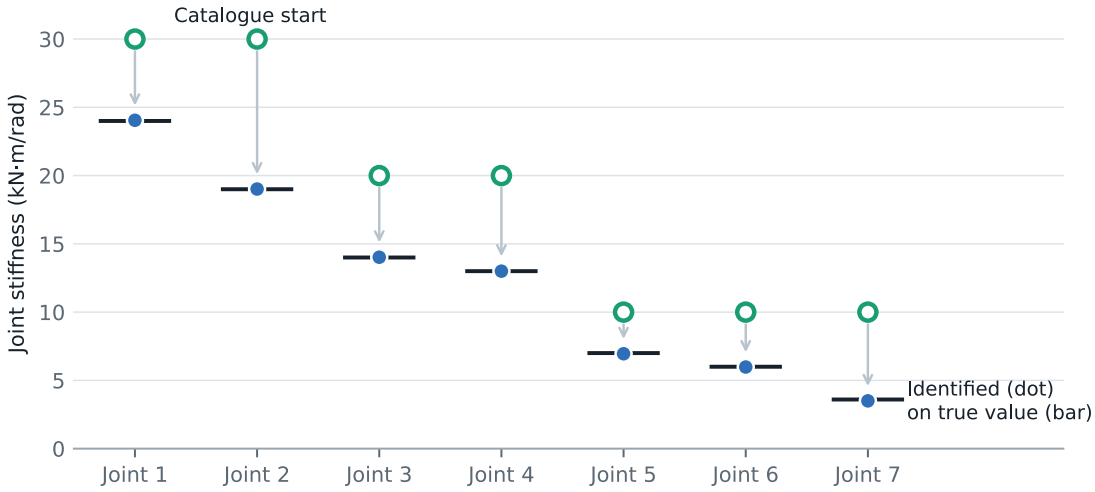


Figure 3. Each joint: catalogue start (open), identified value (filled) and true value (bar).

Prediction on a new motion

A second motion, with different amplitudes and frequencies, was not used in the fit. Against its truth run, the root-mean-square tool error is 2.7 mm for rigid kinematics (largest 4.3 mm), 1.0 mm with catalogue stiffness (1.6 mm) and 1.8 µm with the identified stiffness (3.3 µm), below the noise of the data the model was fitted to.

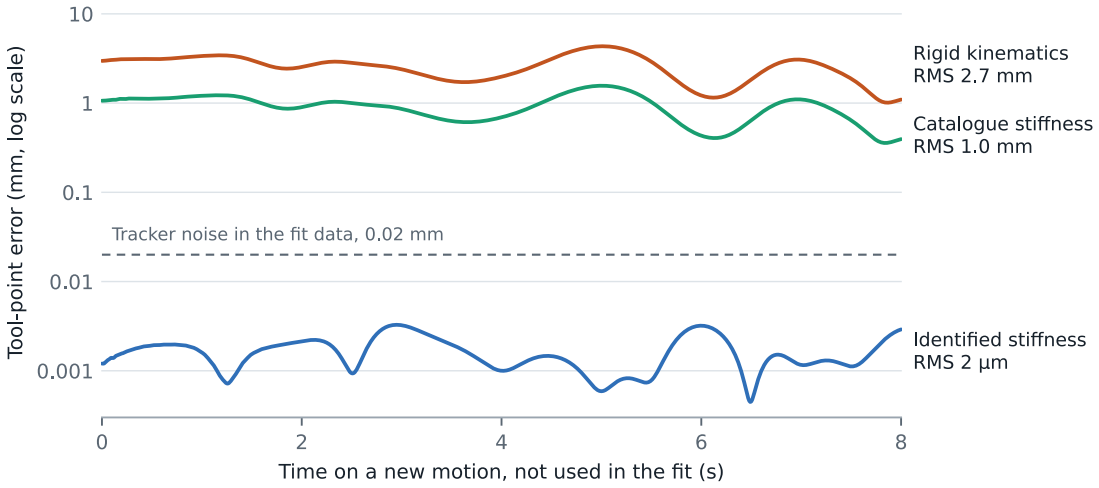


Figure 4. Tool-point error on the new motion, log scale.

What this does and does not show

It shows that the gradients carry through prescribed joint motion, settling and a fit, and that a single dynamic test can pin down every joint, including one a static test cannot see. It does not show accuracy on a real arm. The data come from the same model as the fit, damping is known, and a real arm adds friction, backlash, link flexibility and stiffness that varies with load. One practical lesson: starting the test from the undeflected pose left a 10 Hz mode ringing at about 0.2 mm when the motion began, and that made the fit strongly nonlinear. Letting the arm settle first removed it.

4. Scope

Covered

Forward sensitivities for spring stiffness and damping and the 12 diagonal bushing coefficients, one scalar parameter per sensitivity instance; bushing support is opt-in. The trajectory studies use index-3 HHT with full-rank revolute, spherical and translational joints, rotary motion drivers, and a parameter-independent initial pose and velocity.

Not covered yet

Other prescribed motions (translational, 3D position and orientation), contact, controllers and runtime loads, redundant or rank-changing constraints, projection and other integrators. Adaptive step decisions are not differentiated. The Stewart motion benchmark does not establish differentiability of that whole model.

References

[1] Y. Cao, S. Li, L. Petzold and R. Serban, “Adjoint sensitivity analysis for differential-algebraic equations: the adjoint DAE system and its numerical solution,” *SIAM Journal on Scientific Computing*, vol. 24, no. 3, pp. 1076–1089, 2003.

[2] H. M. Hilber, T. J. R. Hughes and R. L. Taylor, “Improved numerical dissipation for time integration algorithms in structural dynamics,” *Earthquake Engineering and Structural Dynamics*, vol. 5, pp. 283–292, 1977.

[3] D. Negrut, R. Rampalli, G. Ottarsson and A. Sajdak, “On an implementation of the Hilber–Hughes–Taylor method in the context of index 3 differential-algebraic equations of multibody dynamics,” *Journal of Computational and Nonlinear Dynamics*, vol. 2, no. 1, pp. 73–85, 2007.

[4] E. Abele, M. Weigold and S. Rothenbücher, “Modeling and identification of an industrial robot for machining applications,” *CIRP Annals*, vol. 56, no. 1, pp. 387–390, 2007.

[5] C. Dumas, S. Caro, S. Garnier and B. Furet, “Joint stiffness identification of six-revolute industrial serial robots,” *Robotics and Computer-Integrated Manufacturing*, vol. 27, no. 4, pp. 881–888, 2011.

[6] J. Swevers, C. Ganseman, D. B. Tükel, J. De Schutter and H. Van Brussel, “Optimal robot excitation and identification,” *IEEE Transactions on Robotics and Automation*, vol. 13, no. 5, pp. 730–740, 1997.

[7] A. Verl, A. Valente, S. Melkote, C. Brecher, E. Ozturk and L. T. Tunc, “Robots in machining,” *CIRP Annals*, vol. 68, no. 2, pp. 799–822, 2019.
